

과제 #1

- Darknet
 - website: <https://pjreddie.com/darknet/>
 - source code: <https://github.com/pjreddie/darknet>
- YOLOv3 객체검출 딥러닝 네트워크
 - 논문: <https://arxiv.org/abs/1804.02767>
 - 참고영상: <https://www.youtube.com/watch?v=MPU2HistivI>
 - 설치 및 구동 방법: <https://pjreddie.com/darknet/yolo/>

Outdated. 다음 페이지 참고

- YOLOv3 시스템을 Dockerfile로 패키징하여 Docker Hub에 푸시
- 아래 두 개의 URL 가상대학(ecampus)에 제출
 - Dockerfile이 올라간 **GitHub 링크**
 - Docker Image가 올라간 **Docker Hub 링크**

YOLOv3 객체 검출 Tutorial

```
$ git clone https://github.com/pjreddie/darknet
```

```
$ cd darknet
```

```
$ make
```

```
...
```

```
$ wget https://data.pjreddie.com/files/yolov3.weights
```

```
$ ./darknet detector test cfg/coco.data cfg/yolov3.cfg yolov3.weights data/dog.jpg
```

```
...
```

```
Loading weights from yolov3.weights...Done!
```

```
data/dog.jpg: Predicted in 7.535026 seconds.
```

```
dog: 100%
```

```
truck: 92%
```

```
bicycle: 99%
```

```
$ code -r predictions.jpg
```

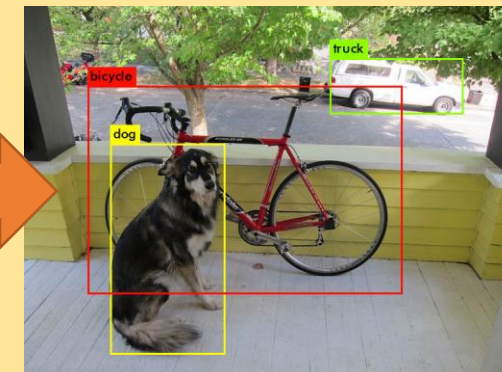
사전 학습된 딥러닝 네트워크 파라미터 파일

VSCode에서 이미지 띄우기

data/dog.jpg



predictions.jpg



기대하는 최종결과물

이미지 URL 입력

```
$ docker run jongchank/yolo https://upload.wikimedia.org/wikipedia/commons/3/3b/SBB Re 450 097 ZKB Nachtnetz.jpg
```

```
Unable to find image 'jongchank/yolo:latest' locally
```

```
latest: Pulling from jongchank/yolo
```

```
...
```

```
Loading weights from yolov3.weights...Done!
```

```
input.jpg: Predicted in 7.633619 seconds.
```

```
train: 100%
```

```
$ docker run jongchank/yolo https://upload.wikimedia.org/wikipedia/commons/4/45/A small cup of coffee.JPG
```

```
...
```

```
Loading weights from yolov3.weights...Done!
```

```
input.jpg: Predicted in 7.909820 seconds.
```

```
diningtable: 94%
```

```
spoon: 98%
```

```
cup: 99%
```



이미지 출력까지는 필요없음